

Re-Solving the Drawing: Physics-Grounded Verification for Text-Guided Editing of Engineering Schematics

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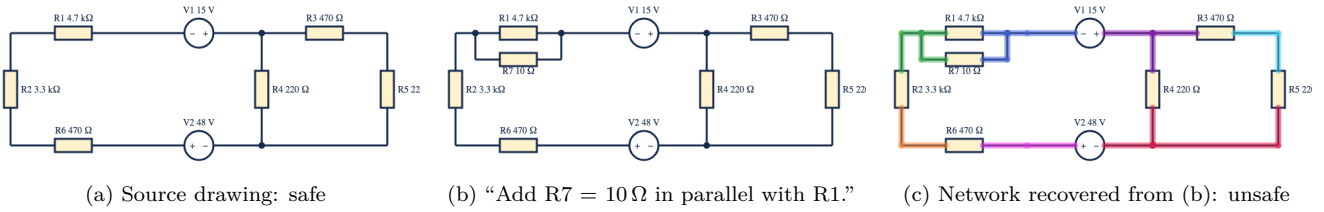


Figure 1. Scoring an edit by re-solving the drawing. A one-resistor edit to a two-source network (a→b) looks innocuous, but it overloads a component elsewhere in the circuit. ReSolve reads only the visible SVG, rebuilding every electrical node from wire endpoints, T-junctions and junction dots (each colour in (c) is one node), reading component values from their labels, and solving the network. Here the edited drawing violates its own note: $R2$ now dissipates 0.84 W against a 0.5 W rating. The same procedure scores any edited drawing, whoever or whatever produced it.

Abstract

Language models now create and edit engineering schematics directly as SVG, but current evaluations score the markup or the pixels, not what the edited drawing means physically. We introduce ReSolve, a verifier that reads a circuit or water-distribution drawing as built. It recovers the network from visible geometry and labels alone, respecting junction and crossing conventions, and re-solves it by nodal analysis or a Newton flow-head solve cross-checked by Hardy Cross. It then grades an edit by whether the drawing describes the intended network and whether that network meets the criteria printed on it. The verifier passes 10 analytical controls and recovers all 10,000 of 10,000 generated test drawings exactly. It is invariant to identifiers, element order and scale. We release 10,000 physics-verified text-guided edits and a harder evaluation tier with up to 16 independent loops. Under this grading, a frontier model (Astra) edits every completed task correctly and analyses circuits exactly, but declares an unsafe pipe network safe without any sign of doubt. Replacing its analysis with a solve of its own drawing removes that error.

1. Introduction

Circuit schematics and water-network plans are routinely exchanged as vector drawings, and language models are now asked to create and edit them directly as SVG markup. An edit that looks correct can still change the physics. Adding a resistor in parallel with one component redistributes current through the whole network (Fig. 1); removing a pipe reroutes flow and can drop the pressure at a distant junction below its service minimum. Whether an edit is acceptable is a property of the network the drawing describes, not of its markup.

Existing evaluations do not measure that property. SVG-editing benchmarks compare the output with a reference by markup or pixel similarity [14, 17], which penalises harmless re-formatting and cannot tell a correctly drawn but dangerous design from a safe one. Engineering-reasoning benchmarks score numerical answers to problems stated in text or code [5, 11, 12], not the drawing a model hands back. Between the two sits the question an engineer actually asks of an edited schematic: if I read this drawing as built, what does it do, and is it safe?

We answer that question mechanically. ReSolve recovers the network from the visible SVG alone: wire

and pipe endpoints, T-junctions, junction dots, component symbols and their labels, read under standard drafting conventions and with no access to identifiers or metadata. It then solves the recovered network by modified nodal analysis for circuits and by a Newton solve of flows and heads for pipe networks, and compares the solution with the intended design. Because the score depends only on the network, any drawing can be graded the same way, whether it comes from a patch, a re-typed SVG or a human drafter.

Contributions.

- A drawing-grounded verifier for resistor and water-distribution networks. It is invariant to identifiers, element order and uniform scaling, it respects crossing and junction conventions, and it is validated against 10 analytical controls and 10,000 generated drawings, every one of which recovers exactly to its source design (Sec. 3).
- Two physics-verified edit datasets: 10,000 text-guided edits over 2,500 networks for training and evaluation, and a harder evaluation-only tier of 400 edits on networks with up to 16 independent loops. Every gold patch reproduces its target exactly, and every safety verdict is computed from the drawing (Sec. 4).
- A measurement of a frontier model (Astra) under this scoring, alongside a small patch-emitting editor paired with the verifier. It shows where the frontier model’s own analysis can be relied upon and where it cannot (Sec. 5).

2. Related work

Vector graphics generation and editing. Learned SVG generators model paths and primitives directly [1] or emit SVG code with a language-model decoder [15]. Instruction-based SVG editing is benchmarked by comparing the edited result with a reference image or markup [14], and scientific-diagram work learns edits from paper revisions [17] or targets structure-faithful generation [10]. These methods treat a drawing as an image or a document. We treat an engineering drawing as a specification of a physical network and score it by what that network does.

CAD and engineering design. Generative CAD models produce parametric construction sequences [9, 19], and agentic systems couple language models to simulators for mechanics [13] and finite-element analysis [4, 20]. Benchmarks such as FEABench [12], FEM-Bench [11] and PDEAgent-Bench [5] evaluate the reasoning or code that sets up a simulation. Our verifier

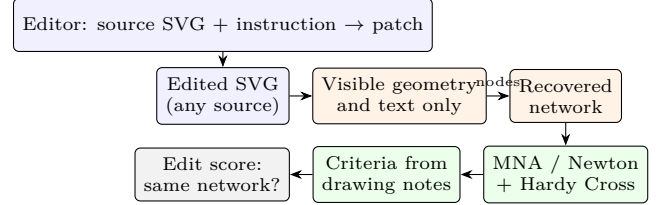


Figure 2. ReSolve. Any edited drawing is read as built: geometry and labels become a network, the network is solved, and the result is checked against the criteria printed on the drawing and compared with the intended design. Identifiers and metadata are never read.

instead reads a finished two-dimensional drawing and reconstructs the model it implies, so it can audit the artefact a designer or a model actually delivers.

Network analysis. Modified nodal analysis [6] is the standard formulation for linear circuits. Pipe networks are classically solved by loop corrections [3] or by the gradient method on flows and heads [18] that underlies EPANET [16]; we use Churchill’s friction factor [2], which is continuous from laminar through fully rough flow. None of this is new. What is new is running these solvers on networks recovered from free-form drawings, and using them as the metric for text-guided editing.

3. Drawing-grounded verification

Task. A source drawing S of a network and an instruction u (e.g. “Add $R7 = 10\Omega$ in parallel with $R1$ ”) define a target design D^* . An editor returns a drawing $\hat{S}$. We ask two questions of $\hat{S}$: does it describe D^* (edit correctness), and does the network it describes satisfy the criteria stated on the drawing (verdict)? Both are decided by recovering a network $\mathcal{R}(\hat{S})$ from the drawing and solving it.

3.1. Recovering a network from a drawing

Recovery parses the SVG with transforms applied and rejects scripts, external references, clipping and masking. It then reads only what is visibly drawn.

Connectivity. Stroked straight segments are candidate conductors (for pipe networks, only heavy strokes). A segment endpoint lying in the interior of another segment is a T-junction and splits it; a filled junction dot splits every segment passing through its centre. Two segments that merely cross stay unconnected, which is the drafting convention for both schematics and piping plans. Endpoints closer than $\tau = 10^{-3}$ drawing units are merged. For circuits, all points joined by wire form one electrical node. For

pipe networks, nodes are points of degree $\neq 2$, dotted points and points on a reservoir outline, and every chain of segments between two nodes is one pipe. A 90° turn along a chain is an elbow; other bend angles are rejected.

Components. A rectangle whose two short-side mid-points coincide with conductor endpoints is a resistor between the corresponding nodes. A circle is classified by where conductors meet it, not by its size: if a conductor passes through its centre it is a junction dot, and if exactly two conductors end on its rim it is a voltage source, whose positive terminal is the one nearer the “+” mark inside it. Filled rectangles touched by pipe ends are reservoirs. Classifying by incidence rather than radius or stroke width keeps recovery unchanged under uniform scaling of the drawing.

Values and criteria. Labels are parsed with fixed grammars (e.g. R3 4.7 kΩ, P5 320 m Ø150, J2 q=6 L/s z=10 m) and assigned one-to-one to the nearest component or pipe, within a reach proportional to the drawing’s own scale (the median resistor body or pipe segment). Design criteria are read from the drawing’s notes: the resistor power rating and supply fuse; the pipe roughness, elbow loss coefficient, minimum pressure head and maximum velocity. A drawing that leaves any component unlabelled, or has two labels competing for one component, is reported as unreadable rather than guessed at.

3.2. Solving the recovered network

Circuits. We use modified nodal analysis [6] with one grounded node per connected part of the network, so a floating fragment carries no current and does not make the system singular. A source whose terminals share a node, or a loop made only of sources, makes the system singular; such a drawing is reported as ill-posed. The outputs are every branch current, every resistor’s dissipation and every source’s current; the criteria are $P_R \leq P_{\text{rated}}$ for each resistor and $|I_V| \leq I_{\text{fuse}}$ for each source.

Pipe networks. For pipe p from node a to node b with flow Q_p , the head loss is

$$h_p(Q_p) = \left(f(\text{Re}_p, \varepsilon/D_p) \frac{L_p}{D_p} + n_p K_{\text{elbow}} \right) \frac{v_p |v_p|}{2g}, \quad (1)$$

where $v_p = 4Q_p/(\pi D_p^2)$, f is Churchill’s friction factor [2] and n_p is the number of elbows counted from

the drawing. We solve jointly for pipe flows and junction heads,

$$h_p(Q_p) = H_a - H_b \quad \forall p, \quad \sum_{\text{in}} Q_p - \sum_{\text{out}} Q_p = q_j \quad \forall j, \quad (2)$$

with reservoir heads fixed, by Newton’s method with backtracking [18], stopping at residuals below 10^{-10} m and 10^{-13} m³/s. Because Churchill’s f tends to $64/\text{Re}$ in laminar flow, h_p has a finite, non-zero slope at $Q_p = 0$ and the Jacobian stays regular when a flow reverses. As an independent check, every network is also solved by Hardy Cross loop corrections [3], with multiple reservoirs joined through a virtual datum node. The criteria are $|v_p| \leq v_{\text{max}}$ for each pipe and $H_j - z_j \geq p_{\text{min}}$ for each junction.

3.3. Scoring an edited drawing

An edit is correct when $\mathcal{R}(\hat{S})$ equals the target network D^* in two senses. The first is topological: we compare, for every node, the multiset of named component terminals it touches, which is a complete invariant of a network with named components, together with every component value and every criterion. The second is physical: the two solutions agree, with currents within 10^{-9} relative plus 1 pA, flows within 10^{-9} m³/s and heads within 10^{-7} m. Markup is irrelevant: a drawing that re-orders elements, renames identifiers or re-formats numbers is credited if it describes the same network, and a drawing that is byte-for-byte plausible but disconnects a wire by 0.01 units is not.

3.4. Validating the verifier

Table 1 lists the analytical controls. They include a balanced and an unbalanced Wheatstone bridge; we initially chose bridge values that happened to balance, which made the “unbalanced” control vacuous and hid a sign error, and we record this as a caution. Beyond the controls, 10,000 independently generated drawings (source designs and all their edits, over two seeds) recover to exactly the network they were drawn from, as do all 20,800 drawings in the released datasets. Unit tests confirm four further properties: recovery is unchanged by stripping identifiers, shuffling elements, translating and uniformly scaling the drawing by $\times 0.25$ to $\times 4$; a T-junction joins, while a crossing joins only when dotted; moving only the “+” mark reverses every branch current; and moving one wire endpoint by 0.01 units breaks the connection. Two of these tests found real defects in the first version. A fixed-radius rule for junction dots misclassified dots once the drawing was scaled up, and an absolute current tolerance misjudged an edit whose two sources exactly cancel.

Control	ReSolve	Reference
series current $V/(R1+R2)$	0.0375	0.0375
parallel current $V(1/R1+1/R2)$	0.174545	0.174545
balanced Wheatstone bridge current	-1.34572e-18	0
unbalanced bridge vs Thevenin	-0.00307525	-0.00307525
superposition of two sources	0.263636	0.263636
two reservoirs vs bisection (m3/s)	0.0324668	0.0324668
laminar loss vs Hagen-Poiseuille (m)	4.17135e-05	4.17135e-05
Churchill vs Colebrook at $Re=1e6$ (relative gap)	1.00497	1
Newton vs Hardy Cross, two loops + two reservoirs (m3/s)	1.21552e-14	0
mass balance: reservoir supply = demand (m3/s)	1.38778e-17	0

Table 1. Analytical controls for the two solvers. References are closed forms, an independent bisection, or a second algorithm.

4. Datasets

Resistor networks. A design starts from a grid of 2×3 to 5×5 nodes. Edges are removed at random while the graph stays connected and bridgeless (a bridge would carry no current), until 2–6 independent cycles remain. Each edge becomes a resistor (E6 series, $10\ \Omega$ – $6.8\ k\Omega$), a plain wire (20%), or one of one or two DC sources (5–48 V, random polarity). Designs are rejected if a resistor is shorted, if sources form a loop, or if no current flows. Notes state a resistor power rating (0.25–2 W) and a supply fuse (0.1–1 A). Drawings use standard symbols, with junction dots at nodes of degree three or more.

Water-distribution networks. A random spanning tree of a 2×3 to 5×5 grid receives one to three extra pipes to close loops. Some dead ends are pruned, and one or two reservoirs attach at the boundary. Grid points of degree $\neq 2$, plus about a third of the rest, become junctions carrying a demand (0–15 L/s) and an elevation (0–30 m). The remaining pass-through points become straight runs or 90° elbows. Pipes have lengths of 100–800 m and nominal diameters of 80–300 mm, and notes state the roughness, elbow loss coefficient ($K = 0.9$), minimum pressure head (10–20 m) and maximum velocity (1.5–2.5 m/s).

Edits. Each design receives four instructions. For circuits these are a resistor value change, a source voltage change, a polarity reversal (two-source designs) or opening a branch (one-source designs, provided the network stays solvable), and adding a resistor in parallel with an existing one, drawn as a tapped branch. For

	Main (v1)	Hard
Source designs	2,500	100
Edits (train/val/test)	8,000/1,000/1,000	0/0/400
Circuit loops	3 (1–6)	11 (8–16)
Pipe-network loops	2 (1–4)	7 (4–9)
Edited designs safe	5,368	118
Verdict flips	780	22
Edits using set_text	5,722	209
Edits using set_attributes	490	18
Edits using remove_element	2,009	82
Edits using insert_subtree	1,779	91

Table 2. Datasets. Loops are independent cycles of the recovered network (median, range). “Verdict flips” are edits that turn a safe design unsafe or the reverse. All patches reproduce their targets and all drawings recover to their designs.

pipe networks they are a diameter change, a demand change, removing a pipe (provided every junction stays supplied and no reservoir is cut off), and adding a pipe between adjacent junctions (or, if none exist, a length change). Target drawings are rendered from the edited design. The gold edit is a tree patch derived between source and target SVG, applied with a validating executor and required to reproduce the target exactly. A row is kept only if both drawings recover to their designs and, for pipes, Newton and Hardy Cross agree to $10^{-9}\ m^3/s$.

Statistics. Table 2 summarises both tiers. The hard tier uses 4×5 to 5×6 grids with 8–16 independent loops for circuits and four to eight added pipes for water networks. It is evaluation-only: no model is trained on networks of that size. Splits are by source design, so the four edits of one drawing never cross splits.

5. Experiments

Frontier model. We evaluate Astra (gpt-6-astra) through the Chat Completions API at medium reasoning effort, with a 32,000-token completion cap, no tools, store=false and one sample per task. Astra works in its natural mode: it receives the drafting conventions, the physical model (fluid properties, friction law, loss conventions) and the criteria in the prompt, and it returns the complete edited SVG followed by a JSON analysis stating the verdict, the violating components and the governing quantities. The edited SVG is scored by ReSolve exactly as any other drawing (Sec. 3), and the analysis is compared with the solver’s. Tasks are the first ten test edits by identifier for every (family, edit type) pair of the main tier (100 tasks), and the first four of the hard tier (40 tasks). The API account ran out of credit during the run. All 79 main-tier and

23 hard-tier tasks that completed are reported. Every circuit task completed. The missing tasks are all pipe networks, which failed with “no credits remaining” (26) or a dropped connection (12), and they are excluded rather than counted as failures.

Our editor. ReSolve pairs the verifier with a small editor: Qwen2.5-Coder-Instruct [8] at 1.5B and 7B parameters, fine-tuned with LoRA [7] (rank 16 on the attention projections) for two epochs on the 10,000-edit training split to map source SVG + instruction to a tree patch. The patch is applied by a validating executor, and the resulting drawing is recovered and solved, so the verdict is computed, not generated. Decoding is greedy. The hard tier is never seen in training.

Metrics. Edit: the edited drawing recovers to the target network. Verdict: the stated pass/fail matches the solver’s. Violations: the set of violating components matches exactly. All: all three at once. For systems whose verdict is computed by ReSolve, the verdict is taken from the solve of the edited drawing, whether or not the edit was correct.

5.1. What Astra gets right

Astra is an accurate editor of these drawings. Every one of its completed edits recovers to the target network, including parallel branches drawn as tapped sub-circuits and pipes added between junctions, and it never disconnected a wire or broke the drafting conventions. Its circuit analysis is exact on every completed task, including hard-tier networks with up to 16 independent loops. We did not find the failure boundary we expected for linear networks: at these sizes, unaided nodal analysis is within reach. Each answer costs \$0.20 on the main tier and \$0.38 on the hard tier on average, and takes a median of 44 s and 101 s respectively, with no truncated responses.

5.2. Where it fails: silently

The only substantive error among Astra’s completed answers is in a nonlinear pipe network, and it is not flagged (Fig. 3). The edit is right, but the analysis declares a failing design safe: it reports 20.04 m of pressure head where the drawing yields 19.23 m against a 20 m minimum, and its flows are wrong throughout the loop that contains the edited pipe. Nothing in the answer distinguishes it from the correct ones: it finished normally, used a similar reasoning budget, and presented its numbers with the same precision. This is the failure mode a verifier exists for. It is rare (1 of 79 main-tier answers), it concentrates where the physics is nonlinear and the design sits near a limit, and it cannot

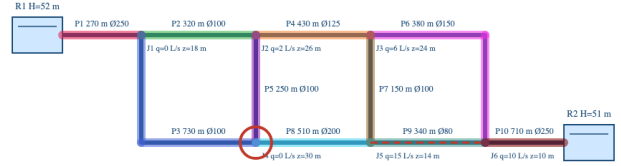


Figure 3. A confident wrong verdict. “Change the length of P9 to 340 m” (dashed) in a two-reservoir network with four loops. Astra’s edit is correct, and it reports the minimum pressure head at J4 (circled) as 20.04 m and the design as passing. Re-solving the drawing gives 19.23 m, below the 20 m minimum stated on the drawing. Several of its pipe velocities are off by up to 67%.

be detected from the answer itself. Replacing Astra’s analysis with the solve of its own drawing removes it (Tab. 3, second row).

5.3. The small editor

[Pending: the id-addressed 1.5B and 7B editors are training.]

5.4. Cost of a verified edit

[Pending.]

6. Limitations and conclusion

Limitations. The networks are procedural: rectilinear layouts, a fixed symbol vocabulary and label grammar, steady DC circuits with ideal sources, and steady incompressible pipe flow with elbow losses only. Recovery is strict by design. A drawing that uses other symbols, draws a resistor as a zig-zag, or places labels far from their components is reported as unreadable, not interpreted. The frontier-model study covers one model at one reasoning effort with one sample per task, and its pipe-network arm is incomplete because the API account ran out of credit, so the hard pipe tier in particular is under-measured. The single wrong verdict is a demonstrated failure mode, not an estimated rate. The numerical tolerance for Astra’s reported velocities gained an absolute floor of 0.01 m/s after we saw that three-decimal rounding of near-zero velocities was being scored as error. Verdicts and edits are unaffected by this change.

Conclusion. An edited engineering drawing can be graded by reading it as built and solving what it describes. This removes the dependence on markup that makes current SVG-editing metrics fragile, and it turns “is this edit safe?” into a computation instead of a claim. Under that grading a frontier model is a strong

System	Edit	Verdict	Violations	All
Main tier (same completed tasks for every system)				
Circuits				
Astra (own analysis)	100 (50/50)	100 (50/50)	100 (50/50)	100 (50/50)
Astra drawing + ReSolve verdict	100 (50/50)	100 (50/50)	100 (50/50)	100 (50/50)
Qwen2.5-Coder 1.5B, untrained			pending	
ReSolve editor 1.5B			pending	
ReSolve editor 7B			pending	
Pipes				
Astra (own analysis)	100 (29/29)	97 (28/29)	97 (28/29)	97 (28/29)
Astra drawing + ReSolve verdict	100 (29/29)	100 (29/29)	100 (29/29)	100 (29/29)
Qwen2.5-Coder 1.5B, untrained			pending	
ReSolve editor 1.5B			pending	
ReSolve editor 7B			pending	
Hard tier (8–16 loops for circuits; unseen in training)				
Circuits				
Astra (own analysis)	100 (20/20)	100 (20/20)	100 (20/20)	100 (20/20)
Astra drawing + ReSolve verdict	100 (20/20)	100 (20/20)	100 (20/20)	100 (20/20)
ReSolve editor 1.5B			pending	
ReSolve editor 7B			pending	
Pipes				
Astra (own analysis)	100 (3/3)	100 (3/3)	100 (3/3)	100 (3/3)
Astra drawing + ReSolve verdict	100 (3/3)	100 (3/3)	100 (3/3)	100 (3/3)
ReSolve editor 1.5B			pending	
ReSolve editor 7B			pending	

Table 3. Edit and verdict accuracy, % (count), on the tasks Astra completed. “Astra drawing + ReSolve verdict” keeps Astra’s edited SVG but replaces its analysis with the solve of that drawing.

ReSolve editor	Edit (circuits)	Edit (pipes)	All
1.5B main test		pending	
1.5B hard tier		pending	
7B main test		pending	
7B hard tier		pending	

Table 4. Our editor on the full test sets: 1,000 main-tier and 400 hard-tier edits.

editor and usually a strong analyst, but its rare analysis errors are silent. Pairing any editor with a drawing-grounded solve makes the verdict exact whenever the edit is right.

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